

MapReduce Algorithm – Part 4

Word Count Problem



Word Count Problem

What is the frequency of each word in the customer feedback?

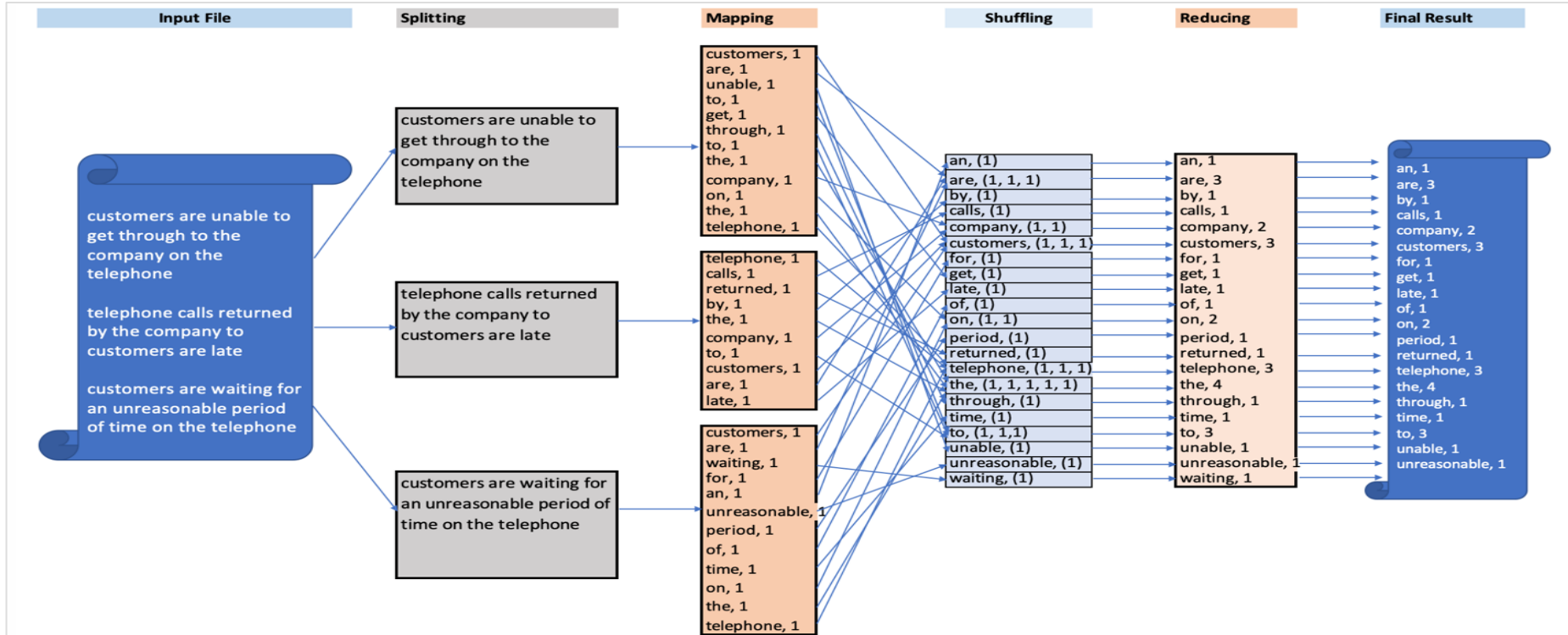
 feedback.txt - Notepad

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```
customers are unable to get through to the company on the telephone  
telephone calls returned by the company to customers are late  
customers are waiting for an unreasonable period of time on the telephone
```



Word Count Problem with Map-Reduce



Sorted Word Count Problem – Part 1

customers are
unable to get
through to the
company on
the telephone

telephone calls
returned by the
company to
customers are
late

customers are
waiting for an
unreasonable
period of time
on the
telephone

MAPPER 1

customers , 1
are , 1
unable , 1
to , 1
get , 1
through , 1
to , 1
the , 1
company , 1
on , 1
the , 1
telephone , 1

telephone , 1
calls , 1
returned , 1
by , 1
the , 1
company , 1
to , 1
customers , 1
are , 1
late , 1

customers , 1
are , 1
waiting , 1
for , 1
an , 1
unreasonable , 1
period , 1
of , 1
time , 1
on , 1
the , 1
telephone , 1

REDUCER 1

an , 1
are , 3
by , 1
calls , 1
company , 2
customers , 3
for , 1
get , 1
late , 1
of , 1
on , 2
period , 1
returned , 1
telephone , 3
the , 4
through , 1
time , 1
to , 3
unable , 1
unreasonable , 1
waiting , 1



Sorted Word Count Problem – Part 2

customers are
unable to get
through to the
company on
the telephone

customers , 1
are , 1
unable , 1
to , 1
get , 1
through , 1
to , 1
the , 1
company , 1
on , 1
the , 1
telephone , 1

telephone calls
returned by the
company to
customers are
late

MAPPER 1

telephone , 1
calls , 1
returned , 1
by , 1
the , 1
company , 1
to , 1
customers , 1
are , 1
late , 1

customers are
waiting for an
unreasonable
period of time
on the
telephone

customers , 1
are , 1
waiting , 1
for , 1
an , 1
unreasonable , 1
period , 1
of , 1
time , 1
on , 1
the , 1
telephone , 1

REDUCER 1

an , 1
are , 3
by , 1
calls , 1
company , 2
customers , 3
for , 1
get , 1
late , 1
of , 1
on , 2
period , 1
returned , 1
telephone , 3
the , 4
through , 1
time , 1
to , 3
unable , 1
unreasonable , 1
waiting , 1

MAPPER 2

1 , an
3 , are
1 , by
1 , calls
2 , company
3 , customers
1 , for
1 , get
1 , late
1 , of
2 , on
1 , period
1 , returned
3 , telephone
4 , the
1 , through
1 , time
3 , to
1 , unable
1 , unreasonable
1 , waiting

REDUCER 2

1 , an
1 , by
1 , calls
1 , for
1 , get
1 , late
1 , of
1 , period
1 , returned
1 , through
1 , time
1 , unable
1 , unreasonable
1 , waiting
2 , company
2 , on
3 , are
3 , customers
3 , telephone
3 , to
4 , the

